National University of Lesotho
Department of Physics and Electronics

EE3304 - Electrical Circuit Analysis
Semester #2 - 2013/2014

Lab 00: Introduction

Due: January 28th, 2014 at 5:00 pm

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1 Lab Brief and Purpose

This lab introduces student to the use of MATLAB for solving electronics problems. MATLAB is a a high-level programming language well-suited for numerical computations. It has a large collection of toolboxes in a variety of application domains. Some examples of these toolboxes are control system, signal processing, neural network, and image processing, etc. A toolbox is a set of functions that can be used for computations in a given application domain. You should complete the pre-lab exercises before coming to the lab.

2 Pre-Lab Exercises

1. Mr Fitz is a farmer. He has both goats and chickens in his farm. In total there are 72 animals and 200 legs. How many chickens does he have? How many goats does he have?
2. Mr Fitz is really a very funny chap (I usually visit his farm just to laugh). On one lazy Sunday he performed the following tasks,
   - He put a bucket of 3 chicken eggs and 2 buck eggs on a scale, and it weighed 130 g.
   - He removed the first bucket from the scale and put another bucket of 5 chicken eggs and 4 duck eggs, and it weighed 240 g.

   How much does a duck egg weigh? how much does a chicken egg weigh?
3. Mr. Fitz goes to the livestock market and wants to sell three sheep and buy two pigs, but is M200.00 short. He then decides to sell two sheep and buy one pig and this time goes home with a M20.00 change. All the sheep have the same selling price and pigs the same purchase price. What is the cost of a pig and price of a sheep?
4. For the circuit shown in Figure 1
   (a) Use Kirchhoff's laws to find i_1 and i_2. [Leave your answer in matrix form]
   (b) Evaluate your answer for the following values:
   $v_s = 5\text{V}$, $R_1 = R_2 = R_C = 1\ \Omega$.
   (c) Evaluate your answer when $v_s = 2\sin(\omega t)$ V.

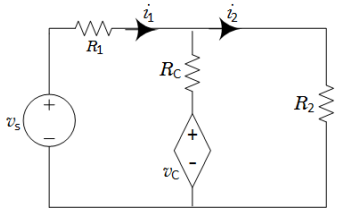


Figure 1: The circuit for Pre-Lab Exercise 4

5. For the circuit shown in Figure 2

   (a) Use mesh analysis to find i_1, i_2, and i_3. [Leave your answer in matrix form]

   (b) Evaluate your answer for the following values:
   $v_1 = 5$V, $v_2 = 14$ V, $i_x = 2$ mA, $R_1 = 5$ kΩ, $R_2 = 3$ kΩ, $R_3 = 4$ kΩ, $R_4 = 2$ kΩ, and $R_5 = 7$ kΩ.

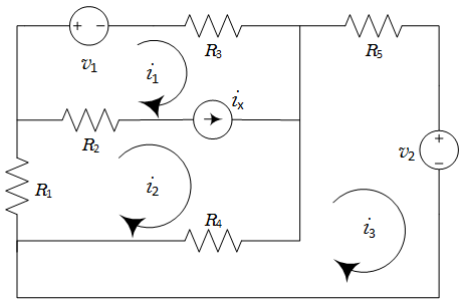


Figure 2: The circuit for Pre-Lab Exercise 5

6. For the circuit shown in Figure 3.

   (a) Use nodal analysis to find the voltages at nodes 1, 2 and 3. [leave your answer in matrix format]

   (b) Evaluate your answer for the following values:
   $v_1 = 5$ V, $v_2 = 10$ V, $i = 3$ mA, $R_1 = 2$ kΩ, $R_2 = 6$ kΩ, $R_3 = 3$ kΩ, $R_4 = 7$ kΩ, and $R_5 = 5$ kΩ, and $R_6 = 4$ kΩ.

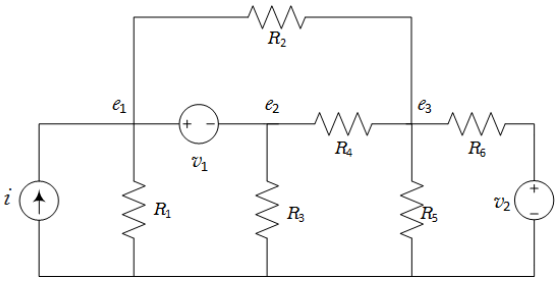


Figure 3: The circuit for the Pre-Lab Exercise 6

3 In-Lab Exercises

1. The code that follows plots 2 functions on the same graph.

```
1   close all
2   clear all
3   clc
4   t = linspace(0,.98,99);
5   y1 = sin(2*pi*4*t);
6   plot(t,y1);
7   y2 = cos(2*pi*4*t);
8   hold on;
9   plot(t,y2, 'r');
10  grid on;
11  legend('sin','cos');
12  print -dpng 'myFirstPlot.png'
```

(a) Create an .m file to implement the above piece of code.

(b) What does the code in line 1, 2 and 3 do? and line 8?

(c) Modify the code to plot the *sin* and *cos* functions in *solid* green and *dashed* blue lines respectively.

(d) Modify the code such that our plot is labelled:

- Title: 'my first plot'
- x-axis: 'time'
- y-axis: 'amplitude'

(e) Modify the axis scale such that we can only see the plot for t = 0 to 0.5.

2. Write a Matlab script that computes a factorial of numbers for $n = 1$ to 12:

$$f_n = n \times f_{n-1}$$

with the initial condition,

$$f_1 = 1$$

3. Write a Matlab function that returns a factorial of a number n.

4. Write down a matrix equivalent of your mathematical equations developed in Pre-Lab Exercise (1), (2) and (3)

$$A\mathbf{x} = \mathbf{b}$$

From linear algebra, the solution of the above equation is,

$$\mathbf{x} = A^{-1} \times \mathbf{b}$$

On the Matlab command prompt, type the following:

```
> A = your 'A' matrix and hit [ENTER] e.g.  A = [a11,a12;a21,a22];
> b = your 'b' vector [ENTER] e.g.  b = [b1;b2];
> x = inv(A)*b [ENTER]
> y = A\ b
```

What can you say about x and y?

5. Refer to your answer in Pre-Lab Exercise 4c. Write Matlab script that plot i_1 and i_2 as a function of time t. [$\omega = 10$ rad/s]

6. Write Matlab script that evaluate your Pre-Lab Exercises 4a, 5a, and 6a using values given in Pre-Lab Exercises 4b, 5b, and 6b respectively. Did you get your pre-lab answers right?

4 Post-Lab Exercise

1. Hand in your Pre-Lab work and In-Lab Work.

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